

Satvik Bagul

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EDUCATION

The University of Massachusetts, Amherst | Amherst, MA

Bachelor of Science in Computer Science | GPA: 3.7/4.0

Awards & Honors: *Dean's List* (Fall 2025), *National and Professional Champion*, ADNOC YasInSchools 4x4InSchools (2024); *Innovator's Award*, First Tech Challenge (2017); *Best delegate*, WINMUN (2024); *Honorable mention*, EISJMUN (2024)

TECHNICAL PROJECTS

Noise, Interactive Gestural Audio Controller | Built for YHack 2026 Spring 2026

- Designed a real-time spatial audio interface using *TouchDesigner* to translate webcam-based hand-tracking gestures into dynamic music modulation.
- Engineered a custom signal processing pipeline using *Math* and *Lag CHOPs* to isolate pinch gestures, normalize input data, and apply temporal smoothing to eliminate tracking jitter.
- Integrated the visual environment with *Ableton Live 12* via the *TDAbleton* package, mapping physical gesture coordinates directly to Auto Filter frequencies for live audio manipulation.

Predictive Project Blueprint System | Built for Hack(Her)-413 Spring 2026

- Built a *full stack SaaS platform* that generates AI-powered project execution blueprints with *work breakdown structures*, *timeline forecasting*, *budget modeling*, and *risk analysis* using probability-weighted scoring.
- Implemented *Monte Carlo simulations* (1000 loops) for timeline predictions with *p10/p50/p90 confidence intervals*.
- Developed with *React*, *FastAPI*, and *MongoDB*. Integrated with *Google Gemini* for blueprint generation. Features GANTT charts, risk heatmaps, and PDF export functionality.

SimplySerenity, AI Therapist | Built for Hack Brown Spring 2026

- Developed an AI-powered virtual therapist using *Java* with an *HTML/CSS* frontend to provide real-time, empathetic mental health support.
- Built on *Google Gemini's API* and utilizes tools from *ElevenLabs* that allow the user and AI to interact verbally.
- Integrated with *Stripe API* to provide easy payment solutions
- Link: simplyserenity.tech

Automated Financial Scraper | Personal Project Winter 2025

- Designed an automated financial analytics tool to extract stock data and generate actionable metrics such as moving averages (helps generate buy/sell signals), volatility, and annualized performance.
- Data extracted using *YFinance API*, GUI built using *Streamlit*, uses *pandas* to compile data, data visualization using *matplotlib*, and storage of data using *CSV files*.

ADDITIONAL EXPERIENCE

President of Student Council | Our Own High School, Al Warqa'a April 2024 - March 2025

- Provided *strategic leadership* to the student body, led council initiatives, and served as the *primary liaison* between students and school administration.

Partnerships Division and Tutor | Instilt Educate Dec 2024 - Jan 2025

- Cultivated partnerships with *10+ educational institutions and key organizations*, including the *British Council (India)* and the *Ministry of Education (Ampara)*, also tutored students from African academic institutions.

SKILLS

- **Programming Languages:** Python, SQL
- **Tools and Technologies:** MS Excel, MS PowerPoint, Canva, Git